

2.2 Group 2

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.2 Group 2
Difficulty	Hard

Time allowed: 50

Score: /36

Percentage: /100

Question 1a

This question is about Group 2 elements.

Barium is silvery white but blackens when exposed to air.

This is due to the formation of barium nitride and one other compound.

Write a balanced symbol equation for the two separate reactions which occur.

Include state symbols.

[2 marks]

Question 1b

Barium can also react with water.

A student reacted 0.20 g of barium with 250 cm³ of water.

i)

Calculate the volume of gas, in cm³, produced in this reaction under standard temperature and pressure.

ii)

Calculate the concentration, in mol dm⁻³ of the barium hydroxide solution produced.

[4 marks]

Question 1c

Describe and explain the trend in the reactivity of Group 2 elements with water.

[5 marks]**Question 1d**

Explain why the reaction of calcium with water is a redox reaction.

Use oxidation numbers in your answer.

[2 marks]**Question 1e**

The trend in solubility and strength as a base can also be observed going down Group 2.

Explain how these two trends are connected.

[3 marks]

Question 2a

This question is about Group 2 hydroxides.

Samples of calcium, strontium and barium are reacted with water to form their hydroxides. The resulting solutions are then filtered to collect the precipitates.

Explain the trend in the expected mass for the precipitates.

[2 marks]

Question 2b

Table 2.1 shows the solubility data for the Group 2 metal hydroxides.

Table 2.1

Group 2 metal hydroxide	Solubility / g dm ⁻³ at 20 °C
Magnesium hydroxide	0.140
Calcium hydroxide	1.730
Strontium hydroxide	17.70
Barium hydroxide	38.90

A student determined that a 50 cm³ solution of an unknown Group 2 metal hydroxide contained 802 mg of the metal hydroxide.

Identify the metal hydroxide in the unknown sample.

Show your working

[2 marks]

Question 2c

Magnesium hydroxide reacts with sulfuric acid.

i)

Write the balanced symbol equation for the reaction between magnesium hydroxide and sulfuric acid. Include state symbols.

[2]

ii)

Barium hydroxide also reacts with sulfuric acid. State any differences in the observations, if any, between the reaction of magnesium hydroxide with sulfuric acid and barium hydroxide with sulfuric acid. Explain your answer.

[3]

[5 marks]

Question 3a

This question is about Group 2 compounds.

i)

Write an equation showing the thermal decomposition of calcium nitrate, $\text{Ca}(\text{NO}_3)_2$.

[1]

ii)

State and explain how the thermal stabilities of the nitrates vary down Group 2.

[3]

iii)

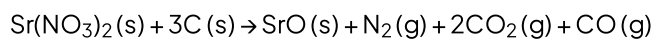
Predict what you would see if a sample of lithium nitrate was heated.

[1]

[5 marks]

Question 3b

The nitrates of calcium, strontium or barium are often added to firework mixtures to produce red or green flames. The equation for the decomposition of one such mixture is as follows.



Calculate the volume of gas given off (measured at room temperature and pressure) when a 10.0 g sample of this mixture decomposes.

Show your working.

[M_r : $\text{Sr}(\text{NO}_3)_2 = 211.6$]

[3 marks]

Question 3c

One of the major ores of magnesium is the mixed carbonate called dolomite, $\text{CaMg}(\text{CO}_3)_2$.

Calculate the percentage loss in mass that would be observed when a sample of dolomite is heated at a high temperature until the reaction had finished.

Show your working

[3 marks]

Model Answers: Hard

1a

a) The balanced symbol equations are:

- $\text{Ba (s)} + \text{O}_2\text{ (g)} \rightarrow 2\text{BaO (s)}$; [1 mark]
- $3\text{Ba (s)} + \text{N}_2\text{ (g)} \rightarrow \text{Ba}_3\text{N}_2\text{ (s)}$; [1 mark]

[Total: 2 marks]

- From being exposed to the air you can deduce that oxidation has occurred so you need to write the equation for the reaction between barium and oxygen
- You might not have come across barium nitride specifically but you can use your Periodic Table to deduce that the charge on the ion is 3-:
 - Group 2 elements have a 2+ charge so for the charges to cancel out the formula must be Ba_3N_2
 - Based on the fact it causes a black coating indicates it is a solid so the state symbol is (s)

1b

b)

i) The volume of gas produced is calculated by:

- $\text{Ba (s)} + 2\text{H}_2\text{O (l)} \rightarrow \text{Ba(OH)}_2\text{ (aq)} + \text{H}_2\text{ (g)}$; [1 mark]
- Mol of Ba = $0.20 \div 137.3 = 0.00146$; [1 mark]
- Volume of H_2 = $0.00146 \times 24\,000 = 35.0\text{ (cm}^3\text{)}$; [1 mark]

ii) The concentration of Ba(OH)_2 is calculated by:

- $\frac{0.00146}{0.250} = 0.00584\text{ mol dm}^{-3}$; [1 mark]

[Total: 4 marks]

- It is common to bring calculations in from other areas of the course to topics such

as Group 2 metals where there is a limited amount of content they can ask about

- In this case, you have not been given the symbol equation for the reaction of barium with water but this is one you are expected to know - you do not need to give state symbols to obtain this mark
- The molar ratio of Ba : H₂ is 1:1
- Make sure you check units, the question has asked for the answer in cm³ not dm³ so you need to multiply by 24 000 not 24
- For part ii) concentration is calculated using $\frac{\text{number of moles}}{\text{volume}}$
- The number of moles of barium hydroxide and barium (calculated in part i) are the same due to the ratio being 1:1
- You must convert the volume from cm³ to dm³ by dividing by 1000

1c

c) Reactivity with water down Group 2:

- Increases; [1 mark]
- The atomic radius increases

OR

There are more shells; [1 mark]

- There is more shielding / screening; [1 mark]
- The nuclear attraction decreases

OR

Increased shielding and distance outweigh the increased nuclear charge; [1 mark]

- It is easier to remove the outer electrons; [1 mark]

[Total: 5 marks]

- This is a common exam question to ask, it could be worded the other way around and you are asked to explain what happens to reactivity going up Group 2

1d

d) This is a redox reaction because:

- Calcium has been oxidised

AND

Its oxidation number increases from 0 (in Ca) to +2 (in Ca(OH)₂); [1 mark]

- Hydrogen has been reduced

AND

Its oxidation number decreases from +1 (in H_2O) to 0 (in H_2); [1 mark]

[Total: 2 marks]

- Remember:**

- The oxidation number of elements is 0
- In this example, the oxidation number of oxygen is -2 and hydrogen is +1
- Make sure you learn the rules for assigning oxidation numbers

1e

e) These two trends are connected because:

- The solubility in water increases as you move down the group

AND

The strength as a base increases as you move down the group; [1 mark]

- Increasing the solubility (in water) means that there are more hydroxide / OH^- ions in solution

OR

Increasing the solubility (in water) increases the concentration of hydroxide / OH^- ions in solution; [1 mark]

- The hydroxide / OH^- ion makes the metal hydroxide act as a base

OR

Base strength is measured by the concentration of hydroxide / $[\text{OH}^-]$ ions; [1 mark]

[Total: 3 marks]

- There is a clear trend in solubility as you descend Group 2;
 - The solubility of the hydroxides of Group 2 **increases as you go down the group**
 - Magnesium hydroxide, $\text{Mg}(\text{OH})_2$, is almost insoluble – it is actually a suspension in water, and not a solution
 - Barium hydroxide, $\text{Ba}(\text{OH})_2$, is the most soluble – it dissolves to produce a strongly alkaline solution
- The strength of a base is measured by the concentration of hydroxide ions

2a

a) The trend in the expected mass for the precipitates is:

- Calcium hydroxide will have the highest / greatest mass (of precipitate) **AND** barium hydroxide will have the lowest / smallest mass (of precipitate)

OR

The mass (of precipitate) will decrease as you move down the group / increase as you move up the group; [1 mark]

- (Because) solubility of Group 2 hydroxides increases going down Group 2; [1 mark]

[Total: 2 marks]

- This question is about the solubility of the Group 2 metal hydroxides in water
 - **Careful:** the talk about precipitates in the question is to distract you
- Calcium hydroxide is sparingly soluble in water so there will be a large amount of precipitate
- The solubility of the hydroxides increases as you move down the group
- Barium hydroxide is very soluble in water so there will be a small amount of precipitate, if any

2b

b) To identify the metal hydroxide in the unknown sample:

- $(802 / 1000) = 0.802 \text{ g (in } 50 \text{ cm}^3)$

AND

$(0.802 \times 20) = 16.04 \text{ g dm}^{-3} \text{ (in } 1000 \text{ cm}^3)$; [1 mark]

- (The unknown metal hydroxide is) strontium hydroxide; [1 mark]

[Total: 2 marks]

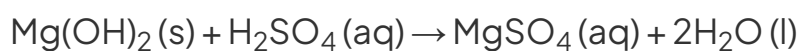
- An answer of strontium hydroxide cannot be achieved without a supporting calculation
- **Careful:** The information in the table wants you to work in g dm^{-3} but you have information in mg and cm^3
 - $1 \text{ g} = 1000 \text{ mg}$
 - $1 \text{ dm}^3 = 1000 \text{ cm}^3$
- To convert the mass from mg to g, divide by 1000
- To scale the mass according to the volume $\times 20$
- Use your answer to identify the correct metal hydroxide from the table by choosing the closest value

- There will always be differences between experimental values and data book tables
 - In this question, the main reasons for the difference could be due to the temperature of the water and impurities in the unknown sample

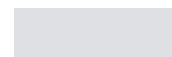
2c

c)

i) The balanced symbol equation for the reaction between magnesium hydroxide and sulfuric acid is:



- Correct formulae; [1 mark]
- Correct balancing; [1 mark]



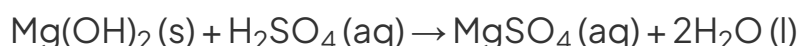
information in mg and cm³

- 1 g = 1000 mg
- 1 dm³ = 1000 cm³
- To convert the mass from mg to g, divide by 1000
- To scale the mass according to the volume x 20
- Use your answer to identify the correct metal hydroxide from the table by choosing the closest value
- There will always be differences between experimental values and data book tables
 - In this question, the main reasons for the difference could be due to the temperature of the water and impurities in the unknown sample

2c

c)

i) The balanced symbol equation for the reaction between magnesium hydroxide and sulfuric acid is:



- Correct formulae; [1 mark]
- Correct balancing; [1 mark]

ii) The differences in the observations between the reactions of magnesium hydroxide with sulfuric acid and barium hydroxide and sulfuric acid are:

- With magnesium hydroxide a colourless solution is produced; [1 mark]
- With barium hydroxide a (white) precipitate is produced; [1 mark]
- Because barium sulfate is insoluble

AND

Magnesium sulfate is soluble; [1 mark]

[Total: 5 marks]

- Group 2 hydroxides are bases so when they react with sulfuric acid they produce a Group 2 sulfate and water
- The solubility of Group 2 sulfates decreases down the group, so magnesium sulfate is soluble - hence it will need the (aq) state symbol in the equation
- Barium is near the bottom of the group so barium sulfate is insoluble and it forms a

white precipitate

- The formation of barium sulfate as a white precipitate should be familiar as the addition of barium ions is a test for the presence of sulfate ions

3a

a) i) An equation showing the thermal decomposition of calcium nitrate, $\text{Ca}(\text{NO}_3)_2$ is:

- $2\text{Ca}(\text{NO}_3)_2 \rightarrow 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$; [1 mark]

ii) Explaining how the thermal stabilities of the nitrates vary down Group 2:

- (Down the group the nitrates) become more stable / are more difficult to decompose / need a higher temperature (to decompose); [1 mark]
- Because the (ionic) radius of the cation / Group 2 ion / M^{2+} increases

OR

The charge density of the cation decreases; [1 mark]

- Thus causing less polarisation / distortion of the anion / NO_3^- / nitrate ion; [1 mark]

iii) When lithium nitrate is heated you would see:

- Brown / orange fumes / gas would be evolved

OR

Glowing splint relights; [1 mark]

[Total: 5 marks]

- When Group 2 nitrates decompose, they produce a Group 2 oxide, nitrogen dioxide and oxygen
- The explanation about the trend in the thermal stability of Group 2 nitrates is worth learning and it can also be used to explain the trend in the thermal stability of Group 2 carbonates
- The observations that would be made when lithium nitrate thermally decomposes arise due to the gases that are evolved in the reaction:
 - Nitrogen dioxide is a brown / orange gas
 - Oxygen would relight a glowing splint

3b

b) The volume of gas given off when a 10.0 g sample of this mixture decomposes is:

Method 1:

- Mass of mixture = $211.6 + (3 \times 12) = 247.6$; [1 mark]
- Moles of mixture = $\frac{10}{247.6} = 0.0404$ (mol); [1 mark]
- No of moles of gas produced = $0.0404 \times 4 = 0.1616$ (mol)

AND

$$\text{Volume} = 0.1616 \times 24 = 3.88 \text{ OR } 3.9 \text{ dm}^3; [1 \text{ mark}]$$

Method 2:

- Mass of mixture = $211.6 + (3 \times 12) = 247.6$; [1 mark]
- 1 mole / 247.6 g of mixture will produce $4 \times 24 = 96 \text{ dm}^3$ of gas; [1 mark]
- 10.0 g of mixture will produce $96 \times \frac{10}{247.6} = 3.88 \text{ OR } 3.9 \text{ dm}^3$; [1 mark]

[Total: 3 marks]

- The mass of one mole of the mixture can be calculated by adding together the molar masses of the reactants:
 - $211.6 + (3 \times 12) = 247.6 \text{ g}$
- One mole of the mixture contains 1 mole of strontium nitrate and 3 moles of carbon and will produce the following:
 - 1 mole of SrO
 - 1 mole of N_2
 - 2 moles of CO_2
 - 1 mole of CO
- N_2 , CO_2 and CO are gases which means that a total of 4 moles of gas are produced for every 1 mole of the reaction mixture
- You would lose a mark for not including carbon in the mass of the mixture - this would give a final answer of 4.54 dm^3 which would score 2 marks
- Either method is equally valid, choose which makes the most sense to you

3c

c) The percentage loss in mass that would be observed when a sample of dolomite is heated at a high temperature until the reaction had finished is:

- $M_r(\text{CaMg}(\text{CO}_3)_2) = 40.1 + 24.3 + 24 + 96 = 184.4$; [1 mark]
- $M_r(2\text{CO}_2) = 2 \times 44 = 88$; [1 mark]

- % loss in mass = $\frac{88}{184.4} \times 100 = 47.7\%$ **OR** 48%; [1 mark]

[Total: 3 marks]

- When Group 2 carbonates undergo thermal decomposition reactions, a metal oxide and carbon dioxide are formed
- In this reaction, both calcium oxide and magnesium oxide will be produced:
 - $\text{CaMg}(\text{CO}_3)_2 \rightarrow \text{CaO} + \text{MgO} + 2\text{CO}_2$
- So, 1 mole of dolomite produces 2 moles of carbon dioxide and 88 grams of carbon dioxide are produced for every 184.4 g of dolomite
- This will be the mass that is lost as carbon dioxide which can be calculated as a percentage of the mass of dolomite

